

Expression of Interest (EOI)

Leveraging Public–Private Rice Partnership Models to Enhance Food and Nutrition Security in Cameroon, Côte d’Ivoire, The Gambia, and Nigeria

Section 0 – Title and contact information

This EOI responds to the Scaling for Impact (S4I) Program AoW 2: Flagship 2.2 – Catalyzing Entrepreneurship to Scale Mechanization and Postharvest Access for Smallholders.

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Key CGIAR Center Partners: AFRICA RICE CENTER (lead) – (*Sali Ndindeng, Francis Nwilene, Erasmus Tang, Elnamaky, Raafat, Prem Bindraban* – with collaboration from CIMMYT (mechanization expertise) - *MARENYA, Paswel Phiri and THIERFELDER, Christian Lutz*).

Non-CGIAR Partners: Government & National Institutions of target countries.

- Cameroun: Ministry of Agriculture and Rural Development (MINADER), Mission d'Étude pour l'Aménagement et le Développement de la Région du Nord (MEADEN), MEADEN, Institute of Agricultural Research for Development (IRAD), Rice Value Chain Development Project Management Unit (RVCDP-MU)
- Côte d’Ivoire: Ministry of Agriculture and Rural Development (MEMINADER), Agence de Developpement Rizicole (ADERIZ), Centre National de Recherche Agronomique (CNRA).
- The Gambia: Ministry of Agriculture, NARI, National Seed Secretariat (NSS), Gambia College of Agriculture
- Nigeria: Federal Ministry of Agriculture and Rural Development (FMARD), Niger State Government, National Agricultural Seed Council (NASC)

Private Sector & Development Partners

- Niger Foods Ltd. (Nigeria) – PPP for seed, mechanization, and offtake markets
- eProd Solutions Ltd. (Regional) – Digital agricultural management and monitoring platform
- MARUO Farms (The Gambia) – Commercial farming and mechanization services
- Women Parboiling Groups & SMEs (Regional) – Processing and value addition
- Seed producers and Processors: Africa-wide Rice Seed and Miller's Network, Dangote Rice, Lagos Produce Project, Tropical General Investments (TGI),

Development & Financing Partners

- African Development Bank (AfDB) – REWARD Program (Regional + National)
- Islamic Development Bank (IsDB) – RVCDP (Cameroon)
- World Bank – VIVA-Bénoué (Cameroon)
- Government funding – Value chain and food security programs

Technical link to other Science Programs: Breeding for Tomorrow, Sustainable Farming, Digital Transformation

Section 1 – Innovation Package, Value Proposition & Alignment

1. Problem to be Solved

West Africa's rice demand is growing rapidly, but Cameroon, Côte d'Ivoire, The Gambia, and Nigeria remain heavily import-dependent¹. Each country faces systemic bottlenecks: low adoption of climate-resilient varieties, weak seed certification and distribution, limited mechanization services, and high postharvest losses exceeding 20–30%². Farmers often plant and harvest late, reducing yields and grain quality. Women and youth, who dominate processing and trading, face barriers to finance, technology, and market entry. Climate variability compounds risks through droughts, floods, and unpredictable rainfall³. These challenges undermine food and nutrition security, create vulnerability to international price shocks, and limit opportunities for inclusive economic growth. Current interventions are fragmented, with pilots not reaching scale, and lack robust traceability and accountability mechanisms. An integrated public–private

¹ Arouna et al. 2021. <https://doi.org/10.1016/j.wdp.2021.100291>

² Ndindeng et al. 2021. <https://doi.org/10.1016/j.rsci.2021.04.001>

³ N'guessan et al. 2025. <https://doi.org/10.1016/j.envdev.2025.101311>

partnership model, reinforced by digital monitoring, is urgently needed to transform the rice value chain into a resilient, competitive system that contributes to food sovereignty.

2. Scaling Ambition

By 2028, the initiative will scale a bundle of proven innovations across four countries to accelerate rice self-sufficiency. The ambition is to reach 1 million smallholder farmers, with at least 40% women and 30% youth. Over 4,000 mechanization SMEs/service providers will be supported, expanding timely ploughing, planting, harvesting, and milling services. Adoption of improved technologies will cover 300,000 hectares, contributing to increased local production and resilience. At least 50,000 new or improved jobs will be created along the value chain. Postharvest losses will be reduced by 25%, ensuring higher quality rice for consumers. In Nigeria, structured offtake agreements with processors (Dangote, Lagos Produce Project, TGI) will absorb 1.75 million tons annually. All farmers and SMEs will be digitally registered and monitored through the eProd platform, enabling real-time traceability and accountability. This ambition directly contributes to CGIAR Scaling for Impact KPIs on adoption, job creation, and healthier diets.

3. Innovation & Value Proposition

The innovation bundle integrates five proven solutions⁴. Seed systems: strengthening breeder, foundation, and certified seed multiplication to expand access to climate-resilient rice varieties. Mechanization service models: enabling youth- and SME-driven hubs for timely land preparation, planting, harvesting, and milling, reducing delays and losses. Digital agricultural management (eProd): registering farmers with GPS mapping, monitoring input distribution, traceability, and market offtake; delivering SMS-based weather, GAP, and early warnings; already tested with over 14,000 farmers under TAAT. Climate-smart GAP and services: providing agronomic and climate data to enhance productivity and resilience. Postharvest innovations: scaling GEM parboilers, solar/rice husk dryers, and modern milling to reduce losses and GHG emissions. This integrated package creates value by addressing bottlenecks from production to consumption, ensuring traceability, reducing imports, and improving competitiveness. It directly benefits farmers, women processors, youth entrepreneurs, and consumers with more resilient, nutritious, and affordable rice.

4. Customers & Alignment

⁴ [All technologies | TAAT e-catalogs](#)

Institutional customers include ministries of agriculture, seed certification bodies, mechanization service providers, processors, and digital companies. End-users are smallholder farmers, women parboilers, youth entrepreneurs, and rice consumers. Governments benefit from reduced imports and enhanced food sovereignty; SMEs gain access to markets and technologies; consumers access safer, acceptable and nutritious rice. The solution aligns with Grand Challenge 2: Deliver Inputs and Services at Scale through Inclusive Market Systems, and with Flagship 2.2 by catalyzing entrepreneurship in mechanization and postharvest services. It also reinforces AfDB's Dakar 2 Compacts, ECOWAS Rice Offensive, and national self-sufficiency targets in all four countries.

5. Evidence of Impact

AfricaRice varieties (NERICA, ARICA, CRAMS, hybrids) are already adopted by over 2 million farmers across West Africa⁵. The GEM parboiling system has improved incomes for 500+ women's groups, raising earnings by 20–30%. Mechanization pilots in Nigeria and Côte d'Ivoire increased yields by 25–40% through timely operations. Digital advisory (RiceAdvice) reached 80,000 farmers, boosting yields by 0.5–1.5 t/ha. Under the AfDB-funded REWARD program, AfricaRice is supporting 13 countries to expand resilient rice production, targeting 53 million tons of paddy by 2030. The Niger Foods PPP in Nigeria mobilized 100,000 farmers, mechanized 50,000 ha in 2024 with the potential to expand to 400,000 ha by 2030, and secured structured offtake for 1.75M tons annually by Dangote Ltd. eProd, deployed in TAAT II, has registered 14,366 farmers, mapped 12,110 ha, documented 17.5M seed volumes, and issued SMS advisories across seven countries. These outcomes demonstrate tested innovations, institutional demand, and readiness for scale beyond pilot sites.

Section 2 – Scaling Pathway, Sustainability, Partnerships & Risks

6. Scaling Pathway & Sustainability

The scaling pathway is built on public–private partnerships that combine government support, private-sector investment, and digital accountability. Governments will enable policies for seed certification, mechanization hubs, and climate services. Private enterprises will deliver certified seed, mechanization services, and processing. SMEs and farmer groups will adopt the innovations as viable business models.

Sustainability rests on four pillars:

1. **Seed systems:** revolving funds and contractual arrangements ensure continuity beyond donor support.

⁵ Arouna et al. 2016. <https://doi.org/10.1016/j.gfs.2017.03.001>

2. **Mechanization:** youth/SME hubs using rental and pay-for-service models generate steady revenue.
3. **Digital services (eProd):** subscription-based platform integrated into national systems for farmer registration, traceability, and monitoring.
4. **Processing and marketing:** structured offtake agreements and PPP investment in processing clusters reduce risks and sustain demand.

In The Gambia, REWARD investments in irrigation, seeds and mechanization will be scaled with eProd's digital monitoring. In Cameroon, VIVA-Bénoué and RVCDP infrastructure will be leveraged to expand mechanization and seed hubs. In Côte d'Ivoire, REWARD Regional and Adaptation provide climate-smart technologies anchored at AfricaRice HQ and M'bé station. In Nigeria, Niger Foods' PPP with AfricaRice expertise and structured market agreements ensure long-term sustainability.

Together, these pathways guarantee financial viability, institutional anchoring, and inclusive benefits for women and youth.

7. Partnerships & Leverage

The initiative builds on significant ongoing investments and strong partnerships.

The Gambia (REWARD-G1): Anchored in the Ministry of Agriculture, NARI, NSS, and Gambia College of Agriculture. Private partners include MARUO Farms and mechanization SMEs. Investments cover irrigation (6000 ha), mechanization centers, and seed delivery systems. The EOI will expand digital monitoring and traceability as well as capacity development.

Cameroon (VIVA-Bénoué and RVCDP): VIVA-Bénoué (World Bank) develops 11,000 ha of irrigated land with a focus on rice. RVCDP (IsDB) targets 160,000 farmers and 210,000 jobs, building processing units, warehouses, and access roads. Anchors include MEADEN, MINADER, IRAD, UNVDA, and the RVCDP-MU. The EOI will layer climate-smart seeds, mechanization hubs, digital systems and capacity development.

Côte d'Ivoire (REWARD Regional & Adaptation): The AfDB-funded REWARD Regional (UA 6M) and Adaptation (UA 7.1M) strengthen seed systems, processing, and climate resilience across 13 countries. Côte d'Ivoire benefits from AfricaRice HQ in Abidjan and M'bé station, ensuring institutional leadership. Anchors include MEMINADER, CNRA, ADERIZ, and USSEM-CI – a member of the Africa-wide Seed and Miller Network.

Nigeria (Niger Foods PPP): Niger Foods, backed by the Niger State Government, operates Nigeria's largest fleet of mechanization equipment, with structured offtake for 1.75M tons



annually. Partners include Dangote, Lagos Produce Project, and TGI. AfricaRice provides varietal development, digital services and backstopping.

Cross-cutting (eProd Solutions Ltd.): Proven under TAAT, eProd provides digital registration, traceability, climate-smart advisories, and inventory tracking. It has onboarded 14,000+ farmers across seven countries, with active pilots in Nigeria and Côte d'Ivoire.

These partnerships ensure co-financing, government commitment, and private-sector leadership, reducing risk and enhancing sustainability.

8. Risks & Bottlenecks

Financial risks: SMEs and farmers face limited access to credit. Mitigation: link with blended finance, guarantee schemes, and revolving funds established under REWARD and Niger Foods.

Policy risks: Weak enforcement of seed certification and mechanization standards. Mitigation: work with NASC (Nigeria), USSEM-CI (Côte d'Ivoire), NSS (Gambia), and IRAD (Cameroon) to strengthen systems, with AfricaRice technical backstopping.

Operational risks: Mechanization services may fragment or face maintenance issues. Mitigation: cluster-based hubs managed by SMEs/youth groups with AfricaRice training.

Climate risks: Droughts, floods, and salinity may reduce yields. Mitigation: REWARD's irrigation and water management investments, plus eProd-enabled climate advisories and early warnings.

Market risks: Import competition may depress prices. Mitigation: structured offtake agreements (e.g., Dangote, Lagos Produce Project, TGI), secure markets for local rice.

Social inclusion risks: Elite capture and gender exclusion. Mitigation: quotas for women/youth participation, targeted training, and eProd and MPRO gender-disaggregated data to ensure equity.

Digital adoption risks: Limited connectivity and digital literacy may constrain platform use. Mitigation: hybrid models combining SMS alerts, field agents, and capacity building.

By embedding risk mitigation in design, the initiative ensures resilience, inclusivity, and effectiveness across diverse production systems.

Section 3 – Responsible Scaling Trajectory & Achievable Scaling Target

9. Readiness & Responsible Scaling

The innovations have undergone scaling readiness assessments and are at SRL 7–9⁶. They have been tested under TAAT, REWARD, RVCDP, and Niger Foods. Responsible scaling is ensured through:

- **Gender and youth targeting:** At least 40% women and 30% youth will be prioritized.
- **Environmental safeguards:** Low-carbon dryers, rice-husk fueled processing systems, water-efficient irrigation, and climate-smart GAPs are embedded.
- **Digital traceability:** eProd registers farmers, tracks inputs, and ensures accountability across the chain.
- **Institutional anchoring:** AfricaRice HQ in Abidjan and national R&D systems (NARI, IRAD, CNRA, NSS, NASC) provide technical support.
- **Market linkage:** Structured offtake agreements in Nigeria and processing clusters in Cameroon, Côte d'Ivoire and The Gambia ensure demand.

Together, these measures demonstrate maturity, demand, and readiness for responsible scaling, while ensuring equity, sustainability, and resilience.

10. Achievable Target, Resources & Activities

By 2028, achievable targets include:

- 1 million farmers adopting innovations, all digitally registered.
- 50,000 jobs created, including mechanization operators, seed producers, processors, and extension agents.
- 300,000 ha under climate-smart practices.
- 25% reduction in postharvest losses.

⁶ [All technologies | TAAT e-catalogs](#)



- Structured offtake of 1.75M tons in Nigeria annually.

Resources:

- AfricaRice expertise in varietal development, seed systems, mechanization, and postharvest innovations.
- National R&D institutions (NARI, IRAD, CNRA, NSS, NASC).
- Private sector (Niger Foods, Africa-wide Seed and Miller's Network, MARUO Farms, eProd Solutions Ltd., processors, and SMEs).
- Financing from AfDB, IsDB, World Bank, EU, IFAD, and private investors.

Activities:

- Breeder and foundation seed production and distribution.
- Establishment of quality declared seeds production clusters managed by the private sector.
- Establishment of mechanization hubs and youth-led service models.
- Deployment of the eProd digital platform for farmer registration, traceability, and climate advisories.
- Training-of-trainers for GAP, mechanization, and digital literacy.
- Installation of GEM parboilers, dryers, and improved milling units.
- Facilitation of policy dialogues and PPP agreements for structured offtake.

These activities ensure a coherent pathway from input delivery to market access.

11. Timeline, Milestones & Monitoring

2026: Partnership agreements signed. Farmer registration launched in all four countries via eProd. Pilot mechanization hubs and seed systems operational. Initial training-of-trainers conducted.

2027: Expansion of mechanization hubs and seed enterprises. eProd monitoring scaled to 500,000 farmers. Processing clusters established in Cameroon and Côte d'Ivoire. First structured offtake agreements operational in Nigeria. Mid-term review conducted.

2028: Consolidation and full-scale rollout. 1 million farmers adopting innovations, 50,000 jobs created, 300,000 – 500,000 ha covered, and postharvest losses reduced by 25%. Traceability mainstreamed with digital dashboards. Structured markets absorbing local rice at scale.

Monitoring:

- AfricaRice will use the AfDB MPRO tool and eProd dashboards for real-time monitoring.
- Indicators disaggregated by gender, youth, and geography.
- Regular surveys and field visits validate adoption and impact.
- Annual reviews with governments, SMEs, and farmer groups ensure adaptive management.

These milestones ensure accountability, learning, and achievement of the scaling targets.

Optional Annexes

Additional References

[All technologies | TAAT e-catalogs](#)

Rice Value Chain Development Project Cameroon: [Rice Value Chain Development Project](#)

VIVA-BENUE Project: [home - vivabenoue](#)

REWARD-GAMBIA: [Barrow launches multi-million dollar agricultural projects](#)